

SRI
SRIMATHI SUNDARAVALLI MEMORIAL SCHOOL
CHENNAI - 600 063
PERIODIC TEST - 1 (2026 - 2027)
MATHEMATICS

CLASS : XII
DATE : 09.06.2026

SET -3

MAX.MARKS : 20
DURATION : 40 mins.

MULTIPLE CHOICE QUESTIONS:

1. Which of the following functions from Z to itself are bijective
(a) $f(x) = x^3$ (b) $f(x) = x+2$ (c) $f(x) = 2x+1$ (d) $f(x) = x^2+x$
2. If the function $f: R \rightarrow A$ given by $f(x) = \frac{x^2}{x^2+1}$ is a surjective, then $A =$
(a) R (b) $[0,1]$ (c) $(0,1]$ (d) $[0,1)$
3. Let $f: R \rightarrow R$ is defined by $f(x) = \frac{x^2-8}{x^2+2}$, then f is
(a) one-one but not onto (b) one-one and onto
(c) onto but not one-one (d) neither one-one nor onto
4. Let $f: R \rightarrow R$ be defined by $f(x) = \frac{1}{x}$, then f is
(a) one-one (b) onto (c) bijective (d) not defined
5. For real numbers x and y , define xRy iff $x-y+\sqrt{2}$ is an irrational number. Then the relation R is
(a) Reflexive (b) symmetric (c) transitive (d) none of these
6. Which of the following is not an equivalence relation on Z ?
(a) $a R b \Leftrightarrow a + b$ is an integer (b) $a R b \Leftrightarrow a - b$ is an integer
(c) $a R b \Leftrightarrow a < b$ (d) $a R b \Leftrightarrow a = b$
7. R is a relation on the set Z of integers and it is given by
 $(x, y) \in R \Leftrightarrow |x-y| \leq 1$. Then, R is
(a) reflexive and transitive (b) reflexive and symmetric
(c) symmetric and transitive (d) an equivalence relation
8. Let $A = \{2, 3, 4, 5, \dots, 17, 18\}$. Let R be the equivalence relation on $A \times A$, Cartesian product of A with itself, defined by $(a, b) R (c, d)$ iff $ad = bc$. Then, the number of ordered pairs of the equivalence class of $(3,2)$ is
(a) 4 (b) 5 (c) 6 (d) 7
9. In the set of Z of all integers, which of the following relation R is not an equivalence relation?
(a) $x R y : \text{if } x \leq y$ (b) $x R y : \text{if } x = y$
(c) $x R y : \text{if } x = y+3$ (d) $x R y : \text{if } x-y \text{ is an even integer}$
10. Let R be a relation on the set N of natural numbers defined by nRm iff n divides m . Then R is
(a) reflexive and symmetric (b) transitive and symmetric
(c) reflexive, transitive but not symmetric (d) an equivalence relation

11. The domain of $\cos^{-1}(x^2 - 4)$ is
 (a) $[3, 5]$ (b) $[-1, 1]$
 (c) $[-\sqrt{5}, -\sqrt{3}] \cup [\sqrt{3}, \sqrt{5}]$ (d) $[-\sqrt{5}, -\sqrt{3}] \cup [\sqrt{3}, \sqrt{5}]$
12. Which of the following is the principal value branch of $\operatorname{cosec}^{-1}$?
 (a) $(-\frac{\pi}{2}, \frac{\pi}{2})$ (b) $[0, \pi] - \{\frac{\pi}{2}\}$ (c) $[-\frac{\pi}{2}, \frac{\pi}{2}]$ (d) $[-\frac{\pi}{2}, \frac{\pi}{2}] - \{0\}$
13. The value of the expression $\tan(\frac{1}{2} \cos^{-1} \frac{2}{\sqrt{5}})$
 (a) $2 + \sqrt{5}$ (b) $\sqrt{5} - 2$ (c) $\frac{\sqrt{5}+2}{2}$ (d) $\sqrt{5} + 2$
14. The value of $\sin^{-1}(\cos \frac{33\pi}{5})$ is
 (a) $\frac{3\pi}{5}$ (b) $\frac{-7\pi}{5}$ (c) $\frac{\pi}{10}$ (d) $-\frac{\pi}{10}$
15. The value of $\tan^{-1}(\tan \frac{7\pi}{6})$ is
 (a) $\frac{\pi}{6}$ (b) $\frac{\pi}{2}$ (c) $\frac{\pi}{3}$ (d) $\frac{7\pi}{6}$
16. $\sin[\cot^{-1}\{\tan(\cos^{-1} x)\}]$ is equal to
 (a) x (b) $\sqrt{1-x^2}$ (c) $\frac{1}{x}$ (d) none of these
17. The number solutions of the equation $\tan^{-1} 2x + \tan^{-1} 3x = \frac{\pi}{4}$ is
 (a) 2 (b) 3 (c) 1 (d) none of these
18. If $\tan^{-1} x = \frac{\pi}{10}$ for some $x \in R$ then the value of $\cot^{-1} x$ is
 (a) $\frac{\pi}{5}$ (b) $\frac{2\pi}{5}$ (c) $\frac{3\pi}{5}$ (d) $\frac{4\pi}{5}$
19. The value of $\cos^{-1}(\cos \frac{5\pi}{3}) + \sin^{-1}(\sin \frac{5\pi}{3})$ is
 (a) $\frac{\pi}{2}$ (b) $\frac{5\pi}{3}$ (c) $\frac{10\pi}{3}$ (d) 0
20. If $\theta = \sin^{-1}\{\sin(-600^\circ)\}$, then one of the possible values of θ is
 (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{2}$ (c) $\frac{2\pi}{3}$ (d) $-\frac{2\pi}{3}$

CLASS: XII

DATE : 16.06.2026

MAX. MARKS: 40

DURATION: 1 hr. 30 min.

SECTION - A

I. Multiple Choice Questions

(10 x 1 = 10)

1. Let R be the relation on the set $A = \{1, 2, 3, 4\}$ given by $R = \{(1, 2) (2, 2) (1, 1) (4, 4) (1, 3) (3, 3) (3, 2)\}$ then,

- (a) R is reflexive and symmetric but not transitive
 (b) R is reflexive and transitive but not symmetric
 (c) R is symmetric and transitive but not reflexive.
 (d) R is an equivalence relation.

2. A mapping $f: N \rightarrow N$, where N is the set of natural numbers is defined as

$$f(n) = \begin{cases} n^2, & \text{for } n \text{ odd} \\ 2n + 1, & \text{for } n \text{ even} \end{cases}$$

for $n \in N$, then f is

- (a) surjective but not injective
 (b) injective but not surjective
 (c) bijective
 (d) neither injective nor surjective

3. Range of the function $f(x) = \frac{x}{1+x^2}$ is

- (a) $(-\infty, \infty)$
 (b) $[-1, 1]$
 (c) $[-\frac{1}{2}, \frac{1}{2}]$
 (d) $[-\sqrt{2}, \sqrt{2}]$

4. The value of $\tan \left\{ \cos^{-1} \left(\frac{-2}{7} \right) - \frac{\pi}{2} \right\}$ is

- (a) $\frac{2}{3\sqrt{5}}$
 (b) $\frac{2}{3}$
 (c) $\frac{1}{\sqrt{5}}$
 (d) $\frac{4}{\sqrt{5}}$

5. The principal value of $\cos^{-1} \left\{ \sin \left(\cos^{-1} \frac{1}{2} \right) \right\}$ is

- (a) $\frac{\pi}{3}$
 (b) $\frac{\pi}{6}$
 (c) $\frac{\pi}{4}$
 (d) $-\frac{\pi}{6}$

6. Let $f(x) = e^{\cos^{-1} \left\{ \sin \left(x + \frac{\pi}{3} \right) \right\}}$. Then $f\left(\frac{8\pi}{9}\right) =$

- (a) $e^{\frac{5\pi}{18}}$
 (b) $e^{\frac{13\pi}{18}}$
 (c) $e^{\frac{-2\pi}{18}}$
 (d) $e^{\frac{-13\pi}{18}}$

7. If A is a matrix of order $m \times n$ and B is a matrix such that AB^T and $B^T A$ are both defined then the order of matrix B is

- (a) $m \times m$
 (b) $n \times n$
 (c) $n \times m$
 (d) $m \times n$

8. If $A = \begin{bmatrix} 2 & -1 & 3 \\ -4 & 5 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 \\ 4 & -2 \\ 1 & 5 \end{bmatrix}$ then

- (a) only AB is defined
 (b) only BA is defined
 (c) AB & BA both are defined
 (d) AB and BA both are not defined

9. If $A = \frac{1}{\pi} \begin{bmatrix} \sin^{-1}(\pi x) & \tan^{-1} \left(\frac{x}{\pi} \right) \\ \sin^{-1} \left(\frac{x}{\pi} \right) & \cot^{-1}(\pi x) \end{bmatrix}$, $B = \frac{1}{\pi} \begin{bmatrix} -\cos^{-1}(\pi x) & \tan^{-1} \left(\frac{x}{\pi} \right) \\ \sin^{-1} \left(\frac{x}{\pi} \right) & -\tan^{-1}(\pi x) \end{bmatrix}$ then $A - B$ is equal to

- (a) I
 (b) 0
 (c) 2I
 (d) $\frac{1}{2} I$

10. Assertion: The domain of the function $\sec^{-1} 2x$ is $(-\infty, -\frac{1}{2}) \cup (\frac{1}{2}, \infty)$

Reason: $\sec^{-1}(-2) = \frac{-\pi}{4}$.

- (a) Both Assertion and Reason are true and Reason is correct explanation of assertion.
 (b) Both Assertion and Reason are true but Reason is not the correct explanation of assertion.
 (c) Assertion is true but Reason is false.
 (d) Assertion is false but Reason is true.

SECTION - B

(3 x 2 = 6)

11. Show that the relation R on $\mathbb{R}$ defined as $R = \{(a, b) : (a \leq b)\}$ is reflexive and transitive but not symmetric.

12. Prove that $3\sin^{-1}x = \sin^{-1}(3x - 4x^3)$, $x \in [-\frac{1}{2}, \frac{1}{2}]$

13. Find the matrix X so that $X \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} = \begin{pmatrix} -7 & -8 & -9 \\ 2 & 4 & 6 \end{pmatrix}$

$$X = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

SECTION - C

14. Check whether the relation S in the set of real numbers $\mathbb{R}$ defined by

$S = \{(a, b) : \text{where } a - b + \sqrt{2} \text{ is an irrational number}\}$ is reflexive, symmetric and transitive.

15. Find the value of $\tan \frac{1}{2} \left[\sin^{-1} \left(\frac{2x}{1+x^2} \right) + \cos^{-1} \left(\frac{1-y^2}{1+y^2} \right) \right]$, $|x| < 1$, $y > 0$ and $xy < 1$.

$$a + b + c = -1$$

SECTION - D

(2 x 5 = 10)

16. Let $A = \{x \in \mathbb{Z} : 0 \leq x \leq 12\}$ Show that $R = \{(a, b) : a, b \in A, |a - b| \text{ is divisible by } 4\}$ is an equivalence relation. Find the set of all elements related to 1. Also write the equivalence class [2]

or

Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{x}{x^2+1}$, $x \in \mathbb{R}$ is neither one - one nor onto.

17. If $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$ then find value of K, $A^3 - 6A^2 + 7A + KI = 0$.

SECTION - E

(2 x 4 = 8)

18. A class teacher is keen to assess the learning of her students the concept of 'relations' taught to them. She writes the following five relations each defined on the set

$A = \{1, 2, 3\}$,

$R_1 = \{(2, 3) (3, 2)\}$

$R_2 = \{(1, 2) (1, 3) (3, 2)\}$

$R_3 = \{(1, 2) (2, 1) (1, 1)\}$

$R_4 = \{(1, 1) (1, 2) (3, 3) (2, 2)\}$

$R_5 = \{(1, 1) (1, 2) (3, 3) (2, 2) (2, 1) (2, 3) (3, 2)\}$

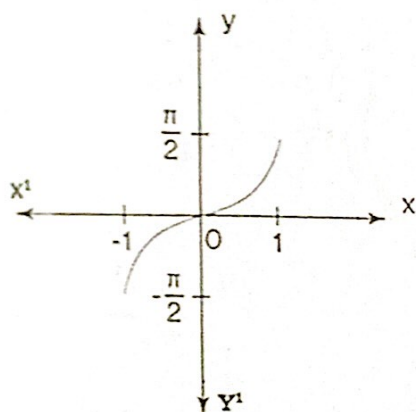
The students are asked to answer the following questions about the above relations:

- (i) Identify the relation which is reflexive, transitive but not symmetric (1)
- (ii) Identify the relation which is reflexive and symmetric but not transitive (1)
- (iii) (a) Identify the relations which are symmetric but neither reflexive nor transitive (2)

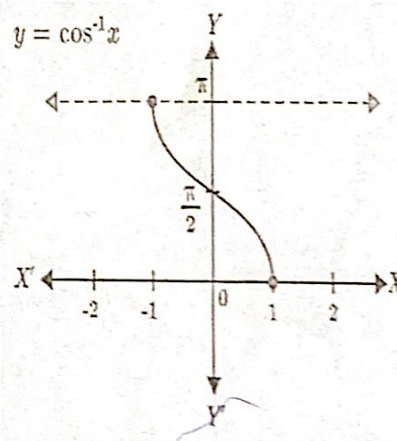
Or

- (b) What pairs should be added to the relation R_2 to make it an equivalence relation?

19. Read the following graph and data, and answer the question given below



Graph of $\sin^{-1}x$



Graph of $\cos^{-1}x$

Function	Domain	Range
$y = \sin^{-1}x$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$y = \cos^{-1}x$	$[-1, 1]$	$[0, \pi]$

- (i) Find the domain of the function if $2 \sin^{-1}x = \sin^{-1}(2x\sqrt{1-x^2})$. (2)
- (ii) Find the domain of the function if $3 \cos^{-1}x = \cos^{-1}(4x^3 - 3x)$. (2)

CLASS: XII

DATE : 21.07.2026

MAX. MARKS: 40

DURATION: 1 hr. 30 min.

SECTION - A

I. Multiple Choice Questions

(10 x 1 = 10)

- If A is a square matrix of order 4 and $|adj A| = 27$, then $A(adj A)$ is equal to
a) 3 b) 9 c) $3I$ d) $9I$
- Value of the determinant $\begin{bmatrix} \cos 67^\circ & \sin 67^\circ \\ \sin 23^\circ & \cos 23^\circ \end{bmatrix}$
a) 0 b) $\frac{1}{2}$ c) $\frac{\sqrt{3}}{2}$ d) 1
- $A = [a_{ij}] = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 3 & 2 \\ 5 & 0 & 4 \end{bmatrix}$ and c_{ij} is the cofactor of element a_{ij} then the value of $a_{21}c_{11} + a_{22}c_{12} + a_{23}c_{13}$ is
a) -57 b) 0 c) 9 d) 57
- If $f(x) = |x| + |x - 1|$, then which of the following is correct?
a) $f(x)$ is both continuous and differentiable at $x = 0$ and $x = 1$.
b) $f(x)$ is differentiable but not continuous at $x = 0$ and $x = 1$.
c) $f(x)$ is continuous but not differentiable at $x = 0$ and $x = 1$.
d) $f(x)$ is neither continuous nor differentiable at $x = 0$ and $x = 1$.
- If $f(x) = \begin{cases} \frac{3\sin\pi x}{5x} & x \neq 0 \\ 2k & x = 0 \end{cases}$ is continuous at $x = 0$ then the value of k is
a) $\frac{\pi}{10}$ b) $\frac{3\pi}{10}$ c) $\frac{3\pi}{2}$ d) $\frac{3\pi}{5}$
- If $x = a \sec \theta$, $y = b \tan \theta$ then $\frac{dy}{dx} = ?$
a) $\frac{b}{a} \sec \theta$ b) $\frac{b}{a} \tan \theta$ c) $\frac{b}{a} \operatorname{cosec} \theta$ d) $\frac{b}{a} \cot \theta$
- A function $f(x) = 10 - x - 2x^2$ is increasing on the interval
a) $(-\infty, \frac{-1}{4}]$ b) $(-\infty, \frac{1}{4}]$ c) $[\frac{-1}{4}, \infty)$ d) $[\frac{-1}{4}, \frac{1}{4}]$
 $-1 - 4x = 0$
 $-4x = 1$
 $x = -\frac{1}{4}$
- A sphere increases its volume at the rate of $\pi \text{ cm}^3/\text{s}$. The rate at which its surface area increases when radius is 1 cm is
a) $2\pi \text{ sq.cm/s}$ b) $\pi \text{ sq.cm/s}$ c) $\frac{3\pi}{2} \text{ sq.cm/s}$ d) $\frac{\pi}{2} \text{ sq.cm/s}$
- The function $f(x) = \frac{x}{2} + \frac{2}{x}$ has a local minimum at x equal to
a) 2 b) 1 c) 0 d) -2
- Assertion:** The absolute maximum value of the function $f(x) = x^5$, $x \in [-1, 1]$ is attained at its critical point $x = 0$.
Reason: For a continuous function on $[a, b]$ absolute maximum occurs either at a critical point or at an end points.
a) Both Assertion and Reason are true and Reason is correct explanation of assertion.
b) Both Assertion and Reason are true but Reason is not the correct explanation of assertion.
c) Assertion is true but Reason is false.
d) Assertion is false but Reason is true.

SECTION - B

(3 x 2 = 6)

11. Find the value of x , if the area of the triangle is 35 sq.cm with vertices $(x, 4)$ $(2, -6)$ and $(5, 4)$

12. Find the value of k for which $f(x) = \begin{cases} \frac{\sqrt{1+kx}-\sqrt{1-kx}}{x}, & \text{if } -1 \leq x < 0 \\ \frac{2x+1}{x-1}, & \text{if } 0 \leq x < 1 \end{cases}$ is

continuous at $x = 0$

13. A particle moves along curve $3y = ax^3 + 1$ such that at a point with x -coordinate 1, y coordinate is changing twice as fast as the x -coordinate. Find the value of a .

SECTION - C

(2 x 3 = 6)

14. Find the intervals in which the function $f(x) = 5x^{\frac{3}{2}} - 3x^{\frac{5}{2}}$ is
(i) increasing (ii) decreasing

15. If $x\sqrt{1+y} + y\sqrt{1+x} = 0$ and $x \neq y$, prove that $\frac{dy}{dx} = \frac{-1}{(x+1)^2}$

SECTION - D

(2 x 5 = 10)

16. Show that the altitude of the right circular cone of maximum volume that can be inscribed in a sphere of radius r is $\frac{4r}{3}$. Also show that the maximum volume of the cone is $\frac{8}{27}$ of the volume of the sphere.

17. (a) Discuss the differentiability of $f(x) = |x-1| + |x-2|$ (3)

- (b) If $y = \sin^{-1}(6x\sqrt{1-9x^2})$, $-\frac{1}{3\sqrt{2}} < x < \frac{1}{3\sqrt{2}}$ then find $\frac{dy}{dx}$ (2)

SECTION - E

(2 x 4 = 8)

18. A carpenter needs to design a wooden box in the shape of a cuboid such that the sum of its length and breadth is 3 cm more than its height. Twice of its length, thrice of its breadth and its height add upto 10 cm. Its breadth added to 7 times its height is 1 cm less than 3 times its length.

On the basis of the above information answer the following questions

- (i) Write the equations representing the various dimensions and express them as the matrix equation $AX = B$ (1)

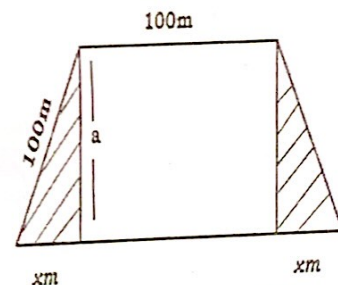
- (ii) Find if A^{-1} exists. Justify your answer (1)

- (iii) (a) Find A^{-1} (2)

(or)

- (iii) (b) Find $A^2 + 7I$ (2)

19. A resort at the top of a hill decided to make a relaxation rectangular field with right triangular fields of equal shape and size, for planting flowers, attached to both sides as shown. They also are thinking of maximizing the total area. The length of rectangle and hypotenuse of right triangular fields are 100m each.



Based on the above information answer the following

- (i) If base of triangular field is x m. Then find width of a rectangular field. (1)

- (ii) Find the perimeter of total enclosed area. (1)

- (iii) (a) Find the total covered area in terms of x . (2)

(or)

- (iii) (b) Find the value of x for which total area is maximum. (2)

**SRI
SRIMATHI SUNDARAVALLI MEMORIAL SCHOOL
CHENNAI - 600 063
I TERM EXAMINATION (2026 – 2027)
MATHEMATICS (041)**

XII-H (14)

CLASS: XII

DATE: 19.09.2026

General Instructions:

MAXIMUM MARKS: 80

DURATION: 3 Hours

Read the following instructions very carefully and strictly follow them:

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) with only one correct option and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided in 2 questions in Section B, 2 questions in Section C, 1 questions in Section D and one subpart each in 3 questions of Section E.
9. Use of calculator is not allowed.

SECTION - A

$$20 \times 1 = 20$$

1. If $f : N \rightarrow W$ is defined as $f(n) = \begin{cases} \frac{n}{2}, & \text{if } n \text{ is even} \\ 0 & \text{if } n \text{ is odd} \end{cases}$ then f is :
 - a) injective only
 - a) a bijection
 - b) surjective only
 - d) neither surjective on injective
2. Let $A = \{3, 5\}$ Then number of reflexive relations on A is.
 - a) 2
 - b) 4
 - c) 0
 - d) 8
3. A relation R is defined on Z as aRb if and only if $a^2 - 7ab + 6b^2 = 0$ Then R is:
 - a) reflexive and symmetric
 - c) transitive but not reflexive
 - b) symmetric but not reflexive
 - d) reflexive but not symmetric
4. The value of $\cos \left[\frac{\pi}{6} + \cot^{-1}(-\sqrt{3}) \right]$ is
 - a) -1
 - b) $\frac{-\sqrt{3}}{2}$
 - c) 0
 - d) 1
5. The domain of the function $\cos^{-1}(4x - 5)$
 - a) $[-1, 1]$
 - b) $(1, 1.5)$
 - c) $(-1, 1)$
 - d) $[1, 1.5]$
6. If matrix $A = [a_{ij}]_{2 \times 2}$ where $a_{ij} = 1$, if $i \neq j$ and 0, if $i = j$ Then A^2 is equal to :
 - a) I
 - b) A
 - c) 0
 - d) none of these

7. If A and B are square matrices of same order such that $AB = A$ and $BA = B$, then $A^2 + B^2$ is equal to :
- a) $A + B$ b) BA c) $2(A+B)$ d) $2BA$
8. Which of the following can be both a symmetric and skew – symmetric matrix?
- a) unit matrix b) diagonal matrix c) null matrix d) Row matrix
9. If M and N are square matrix of order 3 such that $\det(M) = m$ and $MN = mI$, then $\det(N)$ is equal to :
- a) -1 b) 1 c) $-m^2$ d) m^2
10. If A is a square matrix of order 2 such that $\det(A) = 4$ then $\det(4\text{adj}A)$ is equal to
- a) 16 b) 64 c) 256 d) 512
11. If A and B are invertible matrices, then which of the following is not correct?
- a) $(A+B)^{-1} = B^{-1} + A^{-1}$ b) $(AB)^{-1} = B^{-1} + A^{-1}$
c) $\text{adj}A = |A| A^{-1}$ d) $|A|^{-1} = |A^{-1}|$
12. If $f(x) = \begin{cases} 3x - 2, & 0 < x \leq 1 \\ 2x^2 + ax, & 1 < x < 2 \end{cases}$ is continuous for $x \in (0, 2)$ then a is equal to :
- a) -4 b) $\frac{7}{2}$ c) -2 d) -1
13. Let $f(x) = |\sin x|$ $0 \leq x \leq 2\pi$ then
- a) $f(x)$ is discontinuous at 3 points
b) $f(x)$ is differentiable function at infinite number of points
c) $f(x)$ is non-differentiable at 3 points
d) $f(x)$ is non-discontinuous every where
14. If $y = x|x|$ then $\frac{dy}{dx}$ for $x < 0$ is
- a) x b) $-x$ c) $2x$ d) $-2x$
15. The absolute maximum value of function $f(x) = x^3 - 3x + 2$ in $[0, 2]$ is :
- a) 0 b) 2 c) 4 d) 5
16. If $f(x) = 2x + \cos x$ then $f(x)$
- a) has a maxima at $x = \pi$ b) has a minima at $x = \pi$
c) is an increasing function d) is a decreasing function
17. The surface area of a cube increases at the rate of $72 \text{ cm}^2/\text{sec}$. The rate of change of its volume when the edge of the cube is 3 cm is :
- a) $27 \text{ cm}^3/\text{s}$ b) $81 \text{ cm}^3/\text{s}$ c) $72 \text{ cm}^3/\text{s}$ d) $54 \text{ cm}^3/\text{s}$
18. Corner points of the feasible region for an LPP are (0, 2), (3, 0), (6, 0), (6, 8) and (0, 5). Let $F = 4x + 6y$ be the objective function. The minimum value of F occurs at
- a) (0, 2) only
b) (3, 0) only
c) the midpoint of the line Segment joining the points (0, 2) and (3, 0) only
d) any point on the line segment joining the points (0, 2) and (3, 0)

ASSERTION – REASON BASED QUESTIONS:

(Question numbers 19 and 20 carries 1 mark each. Select the correct answer from the options given below)

- a) Both A and R are true and R is the correct explanation of A
b) Both A and R are true. But R is not the correct explanation of (A)
c) (A) is true, but (R) is false
d) (A) is false but (R) is true

19. **Assertion (A)** : Domain of $f(x) = \cos^{-1} x + \sin x$ is $[-1, 1]$

Reason (R) : Domain of a function is the set of all possible values for which function will be defined

20. **Assertion (A)** : In a linear programming problem, if the feasible region is empty, the linear programming problem has no solution.

Reason (R) : A feasible region is defined as the region that satisfies all the constraints.

SECTION - B

5 x 2 = 10

21. Prove that the greatest integer function $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = [x]$ is neither one - one nor onto.

22. Express $\tan^{-1} \left[\frac{\cos x}{1 - \sin x} \right]$ where $-\frac{\pi}{2} < x < \frac{\pi}{2}$ in simplest form

(or)

Simplify: $\sec^{-1} \left[\frac{1}{2x^2 - 1} \right]$ $0 < x < \frac{1}{\sqrt{2}}$

23. If the matrix $\begin{bmatrix} 0 & 6 - 5x \\ x^2 & x + 3 \end{bmatrix}$ is symmetric find the value of x .

24. Find the value of λ if the function $f(x) = \begin{cases} \frac{\sin^2 \lambda x}{x^2}, & \text{if } x \neq 0 \\ 1, & \text{if } x = 0 \end{cases}$ is continuous at $x = 0$

25. Find the value of 'a' for which $f(x) = \sqrt{3} \sin x - \cos x - 2ax + b$ is decreasing on $\mathbb{R}$.

(or)

Find the value of 'a' for which $f(x) = \sin x - ax + b$ increasing on $\mathbb{R}$.

SECTION - C

6 x 3 = 18

26. Let $f: \mathbb{R} - \left\{ \frac{4}{3} \right\} \rightarrow \mathbb{R}$ be a function defined as $f(x) = \frac{4x}{3x+4}$ show that f is a one-one function. Also check whether f is an onto function or not

(or)

Check whether the relation R defined on the set $A = \{1, 2, 3, 4, 5, 6\}$ as $R = \{(a, b); b = a + 1\}$ is reflexive, symmetric or transitive. (2)

27. construct a 3×2 matrix $A = [a_{ij}]$ whose elements are give by $a_{ij} = e^{-ix} \cos \left[\frac{\pi}{2} i + jx \right]$

28. Find $(AB)^{-1}$ if $A = \begin{bmatrix} 1 & 0 \\ -4 & 2 \end{bmatrix}$ and $B^{-1} = \begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}$

29. If $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$ prove that $\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$

(or)

Find the values of a and b if the function $f(x) = \begin{cases} x^2 + 3x + a & x \leq 1 \\ bx + 2 & x > 1 \end{cases}$ is differentiable at $x = 1$

30. If $f(x) = a \log |x| + bx^2 + x$ has extreme value at $x = -1$ and at $x = 2$ then find a and b

31. Solve the following linear programming problem graphically

Maximise $Z = 2x + 3y$

Subject to the constraints

$$x + y \leq 6 \quad x \geq 2 \quad y \leq 3, \quad x, y \geq 0$$

SECTION - D

4 x 5 = 20

32. Show that a function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{2x}{1+x^2}$ is neither one-one nor onto.
Further find set A so that the given function $f : \mathbb{R} \rightarrow A$ becomes an onto function.

33. Use product $\begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix} \begin{bmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{bmatrix}$ to solve the system of equations

$$x + 3z = 9; -x + 2y - 2z = 4 \quad 2x - 3y + 4z = -3$$

34. If $(x-a)^2 + (y-b)^2 = c^2$ for some $c > 0$ prove that $\frac{[1 + (\frac{dy}{dx})^2]^{\frac{3}{2}}}{\frac{d^2y}{dx^2}}$ is a constant independent of a and b.

35. Find the absolute maximum and absolute minimum of function $f(x) = 2x^3 - 15x^2 + 36x + 1$ on $[1, 5]$ (or)

Find the area of the greatest rectangle that can be inscribed in an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

SECTION - E

3 x 4 = 12

36. A company produces three products every day. Their production on certain day is 45 tons. It is found that the production of third product exceeds the production of the first product by 8 tons. while the total production of the first and third product is twice the production of second product.

Based on the above information answer the following questions.

- i) According to given condition write the equations, if x, y and z denotes the quantity of first, second and third product produced. (1)
ii) Write the system of linear equations in matrix form $AX = B$ and hence find $|A|$ (1)
iii) Find A^{-1} (or) Find the values of x, y & z (2)
37. A School is organizing a debate competition with participants as speakers $S = \{S_1, S_2, S_3, S_4\}$ and these are judged by judges $J = \{J_1, J_2, J_3\}$ each speaker can be assigned one judge. Let R be a relation from set S to J defined as

$$R = \{ (x, y) : \text{speaker } x \text{ is judged by Judge } y, x \in S, y \in J \}$$

Based on the above answer the following.

- i) How many relations can be there from S to J? (1)
ii) How many one-one functions can be there from set S to J? (1)
iii) a) A student identifies a function from S to J as $f = \{(S_1, J_1) (S_2, J_2) (S_3, J_2) (S_4, J_3)\}$ (2)
check if it is bijective. (or)
b) Another student considers a relation $R_1 = \{(S_1, S_2) (S_2, S_4)\}$ in set S write minimum ordered pairs to be included in R_1 so that R_1 is reflexive but not symmetric.

38. A tank as shown in the figure below formed using a combination of a cylinder and a cone offers better drainage as compared to a flat bottomed tank. A tap is connected to such a tank whose conical part is full of water. Water is dripping out from a tap at the bottom at the uniform rate of $2 \text{ cm}^3/\text{s}$. The semi vertical angle of the conical tank is 45° .

On the basis of given information answer the following:

- i) Find the volume of water in the tank in terms of radius r. (1)
ii) Find rate of change of radius at an instant when $r = 2\sqrt{2} \text{ cm}$. (1)
iii) Find the rate at which the wet surface of the conical tank is decreasing at an instant when radius $r = 2\sqrt{2} \text{ cm}$. (2)

(or)

- iv) Find the rate of change of height 'h' at an instant when slant height is 4 cm.

